

Problem Statement 1

Operations Visibility Assistant



Northstar Retail Co. is a fictional mid-size retailer shipping from three warehouses (Denver, Columbus, Reno) that grew through acquisitions and never unified its systems. Orders, shipping/fulfillment, support tickets, and inventory all live in separate exports that don't cleanly connect.

The brief candidates receive, from "Priya Chandra, VP of Operations":

Customer complaints about deliveries are up this quarter. We have exports from our systems but no way to see what's going on across them.

Build something that helps our ops team see what's wrong and where to focus — not just another spreadsheet.

That's the whole brief — intentionally incomplete. Candidates get six CSVs (orders, shipping events, support tickets, inventory, customers, products) with real-world data-quality issues baked in: inconsistent warehouse codes, mixed date formats, duplicate rows, mistagged ticket categories, stale inventory data.

The underlying story a strong candidate can surface (not told to candidates): one warehouse had a system migration that caused stale inventory and a spike in delivery delays, which is now escalating into a churn risk with a major B2B account. There's no single correct deliverable — multiple approaches count as "solving" it, as long as the candidate can explain and defend the reasoning behind theirs.

The setup: Candidates play an FDE dropped into a fictional mid-size retailer whose systems (orders, warehouse/shipping, support tickets, inventory) don't talk to each other. They get a deliberately vague brief from a "VP of Operations" — delivery complaints are up, help us see what's going wrong. The data is realistically messy (broken joins, inconsistent formats, mislabeled records) on purpose, and AI tools are explicitly allowed and expected.

What candidates deliver:

A working prototype (their choice of tool/format)

A 5-minute live "client readout" to a judge playing the VP

A one-page "what I'd build next" note

Problem Statement 2

AI-Powered Document Review Assistant for Multi-Department Approval Workflows

Background:

Our organisation processes a high volume of documents — including contracts, business model papers — that require review and approval from multiple departments (Financial, Legal, IT, Audit, etc.). The current workflow relies on Microsoft Word track-changes and comments, circulated manually via email.

Problem:

The existing process creates three core pain points:

- **Version chaos** — When Department A provides comments on v1, the drafter revises to v2, but Department B may still be reviewing v1. Keeping all reviewers aligned on the latest version is manual and error-prone.
- **Comment overload & poor organisation** — Multiple departments leave dozens of comments on the same document. There is no structured way to consolidate, prioritise, or track resolution status across comments from different stakeholders.
- **Slow, repetitive revision cycles** — Drafters spend weeks going back and forth addressing similar or predictable comments that could have been anticipated upfront based on company policies and historical feedback patterns.

Expected Solution Direction:

An AI-powered document review assistant that:

- **Pre-submission Highlight** — Before the document is circulated, the AI scans the draft against company policy documents, compliance requirements, and historical review patterns to **automatically highlight sections that are likely to attract comments or require revision**, giving drafters a chance to fix issues proactively.
- **AI-Suggested Revisions** — For each highlighted section, the system generates a **draft revision suggestion** grounded in company policies and past approved documents, allowing the drafter to accept, modify, or reject the recommendation before submission.
- **Consolidated Comment Management** — Once reviewer comments come in, the AI organises them by department, theme, and priority, and again **suggests response drafts** for each comment to accelerate the revision cycle.

Problem Statement 3

AI-Assisted Personal Tutor (1/2)

Education systems increasingly face a fundamental scalability challenge: every learner has different strengths, weaknesses, learning pace, motivation level, and preferred mode of explanation, yet most classrooms and online learning platforms still provide largely uniform instruction. Teachers and mentors work hard to support individual students, but they often lack sufficient time, real-time visibility, and personalized tools to diagnose each learner's misconceptions, provide timely feedback, and adapt learning materials continuously. As a result, many students fall behind not because they lack ability, but because the right explanation, practice question, or guidance does not arrive at the right moment.

The challenge is to design and build an **AI-assisted personal tutor agent** that can act as an intelligent learning companion for students. The agent should go beyond answering questions. It should understand the learner's goals, assess their current knowledge level, identify gaps and misconceptions, generate personalized learning paths, provide step-by-step explanations, create adaptive exercises, and offer constructive feedback. The tutor should be able to support continuous learning across sessions by remembering progress, adapting its teaching strategy, and helping students build confidence through guided practice rather than simply giving direct answers.

Key Problems to Address

- **Lack of personalization:** Students often receive the same instruction regardless of their background, learning style, or current mastery level.
- **Delayed feedback:** Learners may wait hours or days before receiving meaningful feedback on mistakes, reducing learning momentum.
- **Hidden misconceptions:** Students may appear to complete tasks correctly while still holding incorrect assumptions that affect future learning.
- **Limited teacher bandwidth:** Educators cannot always provide one-to-one support, especially in large classes, remote learning environments, or self-paced programs.
- **Low learner engagement:** Many digital learning tools are static and fail to sustain motivation through interactive dialogue, encouragement, and adaptive challenge levels.
- **Fragmented learning history:** Existing systems often fail to connect past performance, current progress, and future learning recommendations into a coherent learner profile.

Problem Statement 3

AI-Assisted Personal Tutor (2/2)



Expected AI Agent Capabilities

- **Learning diagnosis:** Assess a student's current understanding through conversation, quizzes, assignments, and interaction patterns.
- **Personalized learning plan:** Recommend learning objectives, resources, practice questions, and review schedules based on the student's needs.
- **Adaptive explanation:** Explain concepts at different levels of difficulty using examples, analogies, diagrams in text form, and step-by-step reasoning.
- **Interactive practice:** Generate exercises, hints, worked examples, and follow-up questions that adapt to the learner's responses.
- **Progress memory:** Maintain a structured record of learner progress, recurring mistakes, mastered topics, and recommended next steps.
- **Teacher support:** Provide summaries, learning analytics, and suggested interventions to help educators monitor student progress efficiently.
- **Responsible tutoring behavior:** Encourage learning and reasoning rather than dependence, avoid simply providing final answers, protect student data, and escalate uncertain or sensitive cases appropriately.

Expected Impact

A successful AI-assisted personal tutor could improve learning outcomes by making high-quality individualized support more accessible, especially for students who lack regular access to one-to-one tutoring. It could help students learn at their own pace, receive immediate feedback, and develop stronger metacognitive skills by understanding why they made mistakes and how to correct them. For teachers, the agent could serve as a productivity tool that highlights learning gaps, recommends interventions, and reduces repetitive explanation workload. At scale, such a system could support more inclusive, flexible, and lifelong learning across schools, universities, corporate training, and self-directed education.

Evaluation Criteria

- **Personalization quality:** How effectively the agent adapts explanations, practice, and recommendations to individual learners.
- **Pedagogical effectiveness:** Whether the agent promotes understanding, reasoning, retention, and independent problem solving.
- **Agentic capability:** How well the solution plans, remembers, monitors progress, uses tools, and takes appropriate next actions across learning sessions.
- **Safety and trustworthiness:** Whether the agent provides reliable guidance, acknowledges uncertainty, protects privacy, and avoids harmful or misleading feedback.
- **User experience:** Whether the interaction is engaging, supportive, easy to use, and suitable for students and educators.
- **Scalability and extensibility:** Whether the architecture can support multiple subjects, learner levels, languages, and deployment contexts.